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CSA1306 - TOC 192372014

① TASK 1:- context - free Grammar Desgin and Derivation

context - Free Grammar:- To enforce standarad Mathematical precendence, we structure the gramrcs into - Expression (E), Terms (T), Factor (F)

* Terminals : {+, *, (,), a, b, c}
* Non Terminals: {E, T, F}
* start Symbol: E
* Production Rule:

$E \rightarrow E+T \mid T$

$T \rightarrow T*F \mid F$

$F \rightarrow (E) \mid a \mid b \mid c$

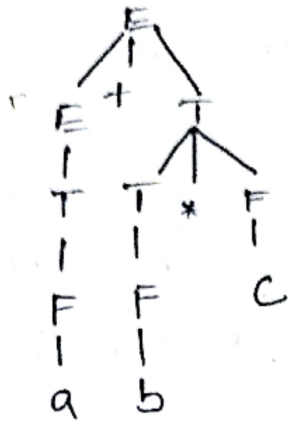
LMD → a+b* c

E → E+T
⇒ T+T
⇒ F+T
⇒ a+T
⇒ a+T*F
⇒ a+F*F
⇒ a+b*F
⇒ a+b* c

RMD → a+b* c

E → E+T
⇒ T+T
⇒ T+F
⇒ T+F*F
⇒ E+T*F
⇒ E+T*c
⇒ E+F*c
⇒ E+b*c
⇒ T+b*c
⇒ a+b*c

Parse Tree:



Task 2: Simplification of Context Free Grammar

Original Grammar:

S → AB | a

A → aA | ε

B → bB | c

C → C

1. Identify and eliminate nullable variable

* Nullable variable : A (Since A → ε)

* Remove A → ε and add productions substituting A with ε where it appears on right hand side

* New Grammar

S → AB | B | a

A → aA | a

B → bB | c

C → C

① Identify non-CNF :- CNF requires all Production to be $V \to VV$ or $V \to T$. The Production $S \to aAB$, $A \to aA$, and $B \to bB$

② Introduce variable for terminal symbol:

$S \to X_a AB \mid b$

$A \to X_a A \mid a$

$B \to X_b B \mid b$

③ Break productions containing more than variable

The production $S \to X_a AB$ has three variables.

$Y \to AB$ to split

$S \to X_a Y$

$Y \to AB$

④ Final chomsky Normal form Grammar:

$S \to X_a Y \mid b$

$Y \to AB$

$A \to X_a A \mid a$

$B \to X_b B \mid a$

$X_a \to a$

$X_b \to b$

2. Identify and eliminate unit productions

* unit Productions : S → B and B → C

* eliminate B → C : Replace it with the right-hand side of C (C → c) . This updates B to B → bB | c.

* eliminate S → B : Replace it with the updated right-hand side of B (bB | c)

* New Grammar:

S → AB | bB | c | a

A → aA | a

B → bB | c

C → c

3. Final Simplified CFG

S → AB | bB | a | c

A → aA | a

B → bB | c

Task 3- Conversion into chomsky Normal Form

Original Grammar:

S → aA | B | b

A → aA | a

B → bB | b

Task 4- Conversion into Greibach Normal form

original grammar

S -> AB | a

A -> aA | a

B -> bB | a

① Identify non GNF Productions:

* A -> aA | a and B -> bB | b already satisfy GNF
* S -> AB violates GNF because it starts with a non-terminal (A).

② final Greibach Normal grammar

S -> aAB | aB | a

A -> aA | a

B -> bB | b

③ Derivation of string

S => aAB

=> aaB

=> aab



Task 5: Context-Free Language Properties and Pumping Lemma

$$L = \{a^n b^n c^n \mid n \geq 1\}$$

1. Determine whether the language is context free

The given language is not context free because it requires equal no of a's, b's and c's

2. State the Pumping lemma for CFL

$$S = uvwxy$$

1. $|vwx| \leq p$
2. $|vx| \geq 1$
3. $uv^i wx^i y \in L$ for $i \geq 0$

$$S = a^p b^p c^p$$

Clearly $S \in L$

$|S| > p$

For $i = 0$ or $i = 2$

$uv^i wx^i y \notin L$

The Pumping lemma condition is violated because the Pumped string does not contain

$L = \{a^n b^n c^n \mid n \geq 1\}$ is not CFL